

## OAKLAND INTL, CA, USA

WMO: 724930

Lat: 37.7444N

Lon: 122.2236W

Elev: 8

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23230

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 36.9       | 39.2 | 20.1                          | 15.1 | 53.0 | 25.8 | 19.8 | 51.6 | 26.5                  | 53.1 | 23.4 | 53.3 | 3.1                   | 50   | 0.521 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 9             | 16.6                   | 83.9            | 64.4 | 79.5 | 63.1 | 75.4 | 62.1 | 66.5                | 78.6 | 65.0 | 75.8 | 63.8 | 73.4 | 10.7                 | 310  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 62.7                            | 85.1 | 69.0 | 61.1 | 80.4 | 67.9 | 59.7 | 76.4 | 66.8 | 31.1          | 78.6 | 29.9 | 75.9 | 29.0 | 73.5 | 73.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |       |            |       |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |       | n=50 years |       |
| Min               |      | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min   | Max        |       |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |
| 23.5              | 20.0 | 18.2 |                            | 32.5 | 93.7 | 2.4                | 3.7                                                | 30.8      | 96.4 | 29.3       | 98.6 | 28.0       | 100.7 | 26.2       | 103.4 |
|                   |      |      |                            | 30.0 | 69.7 | 2.7                | 2.2                                                | 28.0      | 71.3 | 26.4       | 72.6 | 24.9       | 73.8  | 22.9       | 75.4  |
|                   |      |      |                            | DB   |      |                    |                                                    |           |      |            |      |            |       |            |       |
|                   |      |      |                            | WB   |      |                    |                                                    |           |      |            |      |            |       |            |       |

## Monthly Climatic Design Conditions

|       |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|-------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|       |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)   | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 58.4  | 50.6  | 52.7  | 55.4  | 57.4  | 59.9  | 62.6  | 63.8  | 64.6  | 65.0  | 61.9  | 55.5  | 50.7 | (6)  |
| (7)   |                                                                   | DBStd       | 6.60  | 4.07  | 4.10  | 4.13  | 4.33  | 4.12  | 4.12  | 3.38  | 3.61  | 4.52  | 4.42  | 4.55  | 4.45 | (7)  |
| (8)   |                                                                   | HDD50       | 119   | 42    | 18    | 5     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 9     | 44   | (8)  |
| (9)   |                                                                   | HDD65       | 2602  | 447   | 344   | 299   | 232   | 171   | 96    | 64    | 50    | 52    | 117   | 287   | 443  | (9)  |
| (10)  |                                                                   | CDD50       | 3176  | 60    | 94    | 172   | 224   | 308   | 377   | 427   | 453   | 451   | 370   | 173   | 67   | (10) |
| (11)  |                                                                   | CDD65       | 184   | 0     | 0     | 1     | 6     | 13    | 24    | 26    | 38    | 53    | 22    | 1     | 0    | (11) |
| (12)  |                                                                   | CDH74       | 1207  | 0     | 1     | 11    | 53    | 108   | 193   | 125   | 170   | 360   | 178   | 8     | 0    | (12) |
| (13)  |                                                                   | CDH80       | 346   | 0     | 0     | 1     | 9     | 31    | 63    | 31    | 49    | 120   | 42    | 0     | 0    | (13) |
| (14)  | Wind                                                              | WSAvg       | 8.5   | 6.6   | 7.9   | 8.7   | 9.9   | 10.4  | 10.1  | 10.0  | 9.4   | 8.3   | 7.1   | 6.3   | 6.9  | (14) |
| (15)  | Precipitation                                                     | PrecAvg     | 17.9  | 3.7   | 2.8   | 2.8   | 1.2   | 0.2   | 0.1   | 0.0   | 0.0   | 0.2   | 1.3   | 2.7   | 2.6  | (15) |
| (16)  |                                                                   | PrecMax     | 35.5  | 9.1   | 7.2   | 8.0   | 5.4   | 2.4   | 1.2   | 0.7   | 0.9   | 1.5   | 8.2   | 7.9   | 6.1  | (16) |
| (17)  |                                                                   | PrecMin     | 7.2   | 0.3   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.0  | (17) |
| (18)  |                                                                   | PrecStd     | 6.9   | 2.6   | 2.1   | 2.0   | 1.3   | 0.4   | 0.3   | 0.2   | 0.2   | 0.4   | 1.6   | 2.0   | 1.5  | (18) |
| (19)  | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 66.5  | 69.6  | 75.0  | 80.8  | 84.1  | 87.9  | 84.1  | 86.5  | 89.9  | 84.9  | 74.1  | 65.6 | (19) |
| MCWB  |                                                                   |             | 51.3  | 57.3  | 58.2  | 61.5  | 63.0  | 65.3  | 66.7  | 66.2  | 65.5  | 62.1  | 59.0  | 53.1  | (20) |      |
| 2%    |                                                                   | DB          | 61.7  | 65.0  | 69.7  | 73.2  | 75.9  | 79.3  | 77.3  | 78.5  | 82.9  | 79.0  | 69.3  | 62.5  | (21) |      |
|       |                                                                   | MCWB        | 53.9  | 54.8  | 57.4  | 58.8  | 61.0  | 63.4  | 64.6  | 64.6  | 64.5  | 60.4  | 57.2  | 53.9  | (22) |      |
| 5%    |                                                                   | DB          | 59.1  | 62.2  | 65.6  | 68.3  | 70.4  | 73.6  | 72.8  | 73.6  | 77.5  | 74.0  | 65.4  | 60.1  | (23) |      |
|       |                                                                   | MCWB        | 53.1  | 53.8  | 55.8  | 57.4  | 59.5  | 61.6  | 62.6  | 63.3  | 63.2  | 59.6  | 56.5  | 53.9  | (24) |      |
| 10%   |                                                                   | DB          | 57.1  | 59.7  | 62.9  | 64.5  | 66.5  | 70.2  | 70.3  | 71.5  | 72.9  | 70.4  | 63.2  | 57.9  | (25) |      |
|       |                                                                   | MCWB        | 52.3  | 53.7  | 54.9  | 55.9  | 57.8  | 60.5  | 61.4  | 62.4  | 62.4  | 59.1  | 56.0  | 52.9  | (26) |      |
| (27)  | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 58.4  | 59.8  | 61.5  | 63.7  | 66.0  | 68.1  | 68.0  | 67.9  | 68.4  | 65.7  | 62.5  | 59.8 | (27) |
| MCD B |                                                                   |             | 60.5  | 63.8  | 71.5  | 75.6  | 78.9  | 80.7  | 81.4  | 81.6  | 81.8  | 77.0  | 70.1  | 62.0  | (28) |      |
| 2%    |                                                                   | WB          | 56.4  | 57.7  | 59.2  | 60.5  | 62.7  | 64.7  | 64.9  | 65.6  | 66.2  | 63.5  | 60.2  | 57.9  | (29) |      |
|       |                                                                   | MCD B       | 58.8  | 62.2  | 65.6  | 69.3  | 72.5  | 76.3  | 76.1  | 75.5  | 78.1  | 72.9  | 65.5  | 60.1  | (30) |      |
| 5%    |                                                                   | WB          | 54.8  | 56.2  | 57.5  | 58.9  | 60.4  | 62.6  | 63.2  | 64.1  | 64.6  | 62.0  | 58.9  | 56.1  | (31) |      |
|       |                                                                   | MCD B       | 57.2  | 60.5  | 62.9  | 66.2  | 68.5  | 72.7  | 72.1  | 73.0  | 74.8  | 70.7  | 63.4  | 58.5  | (32) |      |
| 10%   |                                                                   | WB          | 53.4  | 54.6  | 56.1  | 57.2  | 58.7  | 60.8  | 61.9  | 62.6  | 63.2  | 60.4  | 57.4  | 54.2  | (33) |      |
|       |                                                                   | MCD B       | 55.9  | 58.5  | 61.3  | 63.5  | 65.8  | 69.3  | 69.7  | 70.3  | 72.0  | 68.1  | 61.7  | 56.7  | (34) |      |
| (35)  | Mean Daily Temperature Range                                      | MDBR        | 13.2  | 13.5  | 14.8  | 14.9  | 13.9  | 15.0  | 13.8  | 14.0  | 16.6  | 17.3  | 15.0  | 13.3  | (35) |      |
| 5% DB |                                                                   | MCD BR      | 16.0  | 17.4  | 20.6  | 22.0  | 21.7  | 22.9  | 18.8  | 18.9  | 27.2  | 25.1  | 19.2  | 15.0  | (36) |      |
|       |                                                                   | MCWBR       | 9.5   | 10.1  | 11.2  | 11.4  | 10.3  | 10.4  | 8.7   | 8.8   | 11.6  | 12.1  | 10.9  | 9.3   | (37) |      |
|       |                                                                   | MCD BR      | 10.8  | 14.2  | 16.3  | 19.1  | 19.1  | 20.7  | 17.8  | 18.7  | 22.2  | 20.0  | 14.7  | 9.4   | (38) |      |
| (39)  |                                                                   | 5% WB       | MCWBR | 7.8   | 9.0   | 9.8   | 10.7  | 9.7   | 10.0  | 8.5   | 8.9   | 10.3  | 10.4  | 9.6   | 7.2  | (39) |
| (40)  | Clear-Sky Solar Irradiance                                        | taub        | 0.324 | 0.330 | 0.342 | 0.360 | 0.364 | 0.360 | 0.369 | 0.349 | 0.348 | 0.340 | 0.324 | 0.325 | (40) |      |
| (41)  |                                                                   | taud        | 2.549 | 2.532 | 2.484 | 2.401 | 2.417 | 2.435 | 2.413 | 2.465 | 2.454 | 2.521 | 2.548 | 2.551 | (41) |      |
| (42)  |                                                                   | Ebn at Noon | 274   | 287   | 292   | 292   | 291   | 291   | 287   | 292   | 286   | 279   | 272   | 265   | (42) |      |
| (43)  |                                                                   | Edh at Noon | 24    | 28    | 32    | 37    | 37    | 36    | 37    | 34    | 32    | 28    | 24    | 23    | (43) |      |
| (44)  | All-Sky Solar Radiation                                           | RadAvg      | 754   | 1054  | 1491  | 1950  | 2253  | 2466  | 2339  | 2086  | 1763  | 1297  | 892   | 662   | (44) |      |
| (45)  |                                                                   | RadStd      | 107   | 128   | 134   | 155   | 145   | 104   | 77    | 88    | 66    | 82    | 57    | 97    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (44 neighbors) | N/A   | N/A     | -2.14  | N/A     | N/A   | N/A   | N/A         | -198  | +243  | N/A   |

Nomenclature: See separate page